YoloOW: A Spatial Scale Adaptive Real-Time Object Detection Neural Network for Open Water Search and Rescue From UAV Aerial Imagery

Jianhao Xu, Xiangtao Fan, Hongdeng Jian, Chen Xu, Weijia Bei, Qifeng Ge, and Teng Zhao

***Abstract*— Personnel and boat detection in unmanned aerial vehicles (UAVs) imagery plays a crucial role in open water search and rescue missions. The diverse perspectives and altitudes of UAV images often result in significant variations in the imagery's appearance and dimensions of personnel and boats, and the false detections arising from water surface flares are acknowledged as a great challenge as well. The existing deep learning-based detection methods employ convolutional blocks with fixed kernel sizes to extract features from the imagery at a fixed spatial scale, which will lead to missed and false detections and severely affect detection accuracy when there are substantial differences in the appearance and size of the target objects. In this article, a spatial scale adaptive real-time object detection neural network, namely YoloOW, was proposed to tackle the challenge of personnel and boat detection amid the diverse UAV imagery, which comprises a feature extractor, a feature enhancer, and a postprocessor. The OaohRep convolutional block was proposed as a pivotal component in constructing the YoloOW and applied to the feature extractor and the feature enhancer. Compared with general convolution blocks, the OaohRep convolution block can extract image features across a wide range of spatial scales, show better scale adaptability, and achieve faster detection speed due to its unique merged convolution layer design. OaohRepBi-path aggregation network (PAN) was proposed in the feature enhancer, which imitated the architecture of the classic algorithm scale invariant feature transform (SIFT) and was successfully applied to deep learning models, showing better scale adaptability. A novel UAV detection box filter (UDBF) module was proposed in the postprocessor, which can effectively remove false detections caused by water surface flares. Experimental results demonstrate that our YoloOW model achieves 37.18% mean averaged precision (mAP) on the SeaDronesSee dataset, surpassing the baseline by 8.43%. This notable improvement positions our model at the first of the leaderboard. The code will be available at https://github.com/Xjh-UCAS/YoloOW.**

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I. Introduction

OPEN water search and rescue is the final line of life safety. The crux of open water search and rescue lies in the rapid detection of distressed individuals in the open water and the precise estimation of their relative positions relative to the rescue boat. At present, almost all open water search and rescue activities need human participation, and the search and rescue efficiency is low [1].

Due to the advancements in oblique photogrammetry and remote sensing, unmanned aerial vehicles (UAVs) have emerged as a promising remote sensing platform, offering advantages, such as high mobility, flexible viewing angles, and cost-effectiveness [2], [3]. UAVs are demonstrating their potential for open water search and rescue.

The primary focus of integrating UAVs into open water search and rescue operations is to achieve automated object detection within UAV aerial imagery [4]. Object detection refers to the process of identifying and locating specific objects within an image or video. In the context of this scenario, it involves identifying elements, such as swimmers, floaters, boats, or other pertinent items present in open water scenes captured by UAVs, which are annotated using bounding boxes. Deep learning-based object detection involves establishing the correlation between image features and bounding boxes during the training phase and during the detection phase generating bounding boxes as an output based on input imagery. Object detection based on deep learning architecture has achieved remarkable breakthroughs and has become the mainstream approach in various applications. For typical scenes, such as microsoft common objects in context (MS COCO) [5] and pattern analysis, statical modeling and computational learning visual object classes (PASCAL VOC) [6], general object detectors [7], [8], [9] have demonstrated strong performance. However, when these detectors are applied to UAV aerial imagery of the open water area, they encounter challenges related to scale variation and water surface flares, which are prevalent issues in UAV aerial imagery of the open water area. Fig. 1 illustrates a few challenges in UAV aerial imagery of the open water area compared to ground-level camera images.

The changes in the altitude and viewing angle of UAVs can result in variations in the pixel size and shape of imaged objects. When UAVs operate at high altitudes, the reduced

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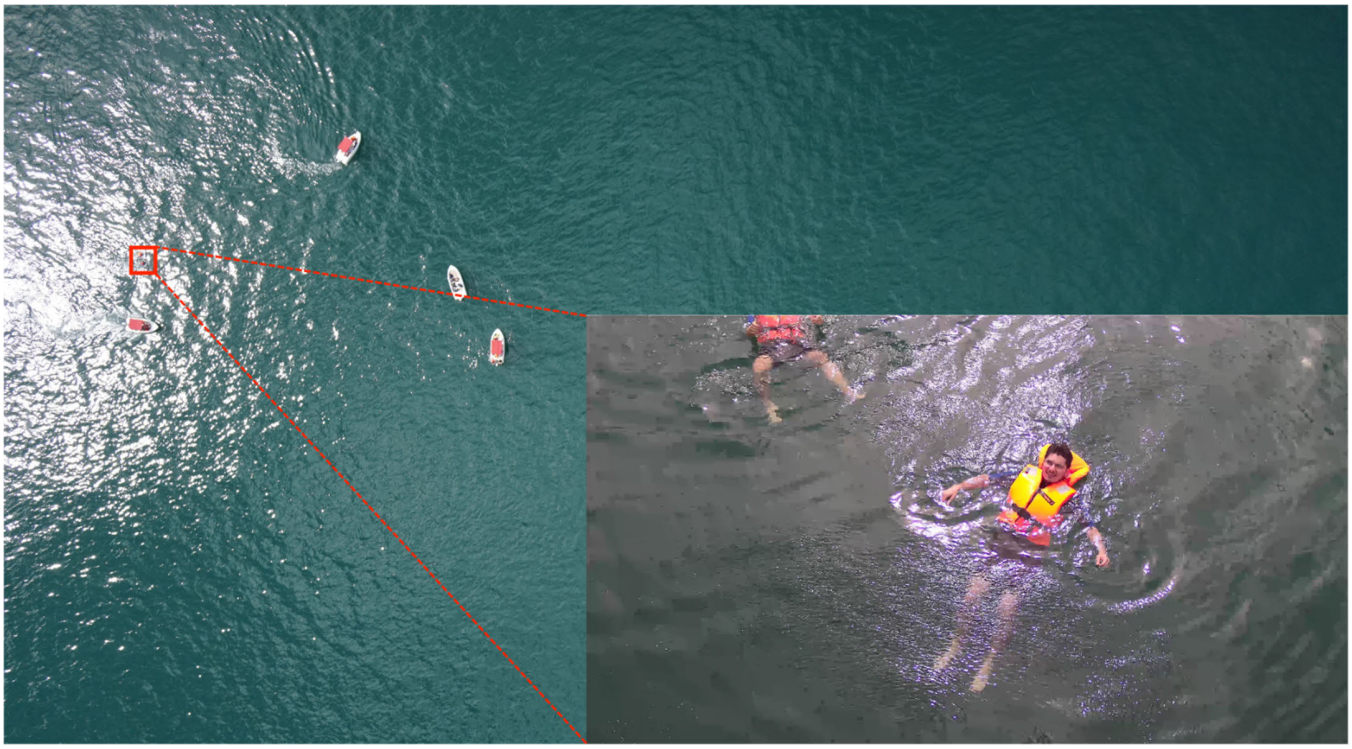


Fig. 1. Aerial images of a consistent open water zone at varied altitudes via UAV. As shown in these two aerial images, variations in UAV altitude are found to induce pronounced alterations in "floater" pixel size. Notably, water surface flares attributed to specular reflections are consistently observable across both images.

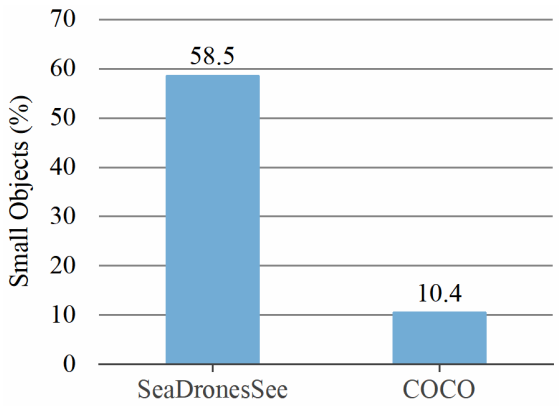


Fig. 2. Comparison of the small objects' frequency among natural scenes and UAV scenes.

pixel size of imaging objects poses significant challenges for general detectors in classifying and localizing diminutive objects [4], [10], [11]. The comparison in Fig. 2 shows that the proportion of small objects in UAV open water search and rescue scenes, such as the SeaDronesSee dataset, happens about five times more than that in natural scenes, such as the COCO dataset. Despite recent advancements in UAV target detection research aimed at enhancing detector robustness against minute object detection challenges, these studies typically involve augmenting small-scale image feature space prediction heads onto existing detectors [12], [13] or added attention modules [14], [15]. The aforementioned methods not only offer limited improvements in accuracy but also lead to an increase in model parameters and floating-point operations, which is disadvantageous for deployment on UAVs.

Furthermore, the phenomenon of water surface flare has persistently posed a challenge for visible light remote sensing in open water areas. Diffuse reflection accounts for only a tiny part of water surface flares compared to common image highlights. This invalidates the real-time image processing algorithm that uses the principle of retaining diffuse reflection and removing specular reflection [16], [17], [18]. In object detection, water surface flares can interfere with all existing detectors, leading to false detection occurrences. Recent studies have neglected to address the interference caused by the water surface flares in UAV aerial imagery of the open water area while conducting object detection [19], [20], [21].

Based on the above challenges faced in open water search and rescue operations, a novel spatial scale adaptive real-time object detection neural network named YoloOW is proposed to improve detection performance in UAV aerial imagery of the open water area. Extensive experiments and complete evaluations demonstrate that our proposed method can solve the false detections caused by the water surface flare challenge and is more effective and faster than the state-of-the-art models. Our method performs best on the SeaDronesSee dataset and provides new insights into detectors in UAV aerial imagery of the open water area. The main contribution of this article is summarized as follows.

1) A novel YoloOW was proposed to address the challenges faced by UAV imagery object detection in open water scenarios.

2) The OaohRep block was proposed to provide YoloOW with better scale adaptability, better fitting ability, and faster detection speed than the general convolution block.

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3) A novel architecture termed dynamically adaptive reparameterized efficient layer aggregation networks (DELANs) and its enhanced variant E-DELAN were proposed to augment the learning capability during model training and diminish floating point operations in the model's inference phase.

4) A UAV detection box filter (UDBF) was proposed to address the water surface flares challenge, which leads to some false detection boxes in UAV aerial imagery of the open water area object detection.

II. Related Work

A. Object Detection in UAV Imagery

Object detection in UAV imagery is essential for open water search and rescue and so on [22] and is more challenging than the natural images. Generic detectors consist of anchor-free detectors [23], [24], [25], and anchor-based detectors [23], [26], [27] perform poorly because of scale variability of objects. Feature pyramid structure [28] fuses geometric information of shallow features and semantic information of deep features to solve scale variability, and both progressive asymmetric feature pyramid network (PAFPN) [29] and bidirectional feature pyramid network (BIFPN) [30] propose different structures for a better fusion of feature maps of multiple stages. Numerous scholarly investigations [13], [31], [32] have unequivocally affirmed that the accuracy of object detection on UAV imagery can be improved by fusing the shallower geometric information of the feature pyramid and increasing the shallow output layer. Furthermore, Wang et al. [33] propose one spatial attention for multiscale feature refinement for object detection in UAV images. The aforementioned approaches have been focused on investigating improved feature enhancers for UAV object detection models. However, they do not account for the fixed scale and number of feature maps used as inputs to the feature enhancers. This limitation constrains the detection accuracy of the models when it comes to objects with varying scales.

B. Object Detection in Open Water

Recently, the deep learning model for object detection in open water has been continuously developed and has produced significant detection results [34], [35], [36]. Numerous scholarly investigations explore object detection models in open water from two perspectives.

Path aggregation network (PAN) [29] transfers shallow geometric information from bottom to top to the overlying feature map to enhance the semantic information of the overlying feature map. By adding a level to the PAN, the challenges associated with multiform and multiscale water surface objects can be mitigated [37]. Its limitation lies in increasing the complexity of the network and reducing the speed of object detection to a certain extent. Motivated by RepBi-PAN [9], OaohRepBi-PAN as a part of feature enhancer for YoloOW is proposed. OaohRepBi-PAN, based on feature fusion, augments geometric information without bringing in excessive computation burden.

Many researchers improve the performance of the object detection model by inserting a convolutional attention module [38]. By enhancing the recurring features within the feature map, the proposed method demonstrates a substantial improvement in accuracy within scenes characterized by densely distributed objects [39]. However, its effectiveness diminishes in scenarios where objects are sparsely distributed [22]. Moreover, this approach also leads to an exponential increase in model parameters and training costs.

None of the aforementioned studies addressed the influence of the water surface flare phenomenon on object detection tasks. For object detection in the open water environment, the phenomenon of water surface flare is inevitable [40], and the proportion of specular reflection in radiation is much higher than that of diffuse reflection [41]. The existing specular highlight removal algorithm is rendered ineffective in this context, necessitating an alternative approach to address the issue of false detection boxes caused by water surface flare [16], [17], [18]. Consequently, our focus shifts toward exploring potential solutions during the postprocessing stage.

III. Method

As shown in Fig. 3, the proposed YoloOW model consists of the following four key parts: OaohRep block, DELAN and E-DELAN, OaohRepBi-PAN, and UDBF. The proposed methodology employs the OaohRep block and the DELAN module within the feature extractor to effectively extract adaptive spatial scale features from UAV imagery. The extracted features are subsequently provided as an input to the feature enhancer. In the feature enhancer, a structure akin to a Gaussian difference pyramid called OaohRepBi-PAN is proposed. By integrating features across layers and scales, OaohRepBi-PAN achieves robustness against scale variations in UAV imagery. Subsequently, the fused features are converted into bounding boxes using the IDetect module, which outputs the bounding boxes to the postprocessor for further analysis and processing. The postprocessor receives an extensive set of low-confidence redundant bounding boxes surrounding the target object, along with false detection boxes resulting from water surface glare as an input. To effectively eliminate the low-confidence redundant bounding boxes, the nonmaximum suppression (NMS) module was utilized. Furthermore, by incorporating drone metadata as an input, the UDBF module successfully alleviates false detections caused by water surface flares.

A. OaohRep Block

This section introduces the OaohRep block that consists of OaohRepConv and RepConv, which are used to generate feature maps with adaptive spatial scale, and this module structure is illustrated in Fig. 4(a).

The convolutional block, as the fundamental component of convolutional neural networks, significantly influences the accuracy and speed of detection models. Previous studies have demonstrated that multibranch convolutional blocks, as presented in [42], [43], [44], and [45], often exhibit superior classification performance compared to single-path convolutional blocks [46], [47]. However, this improvement in classification performance is often accompanied by reduced parallelism and increased inference latency. Conversely, plain single-path

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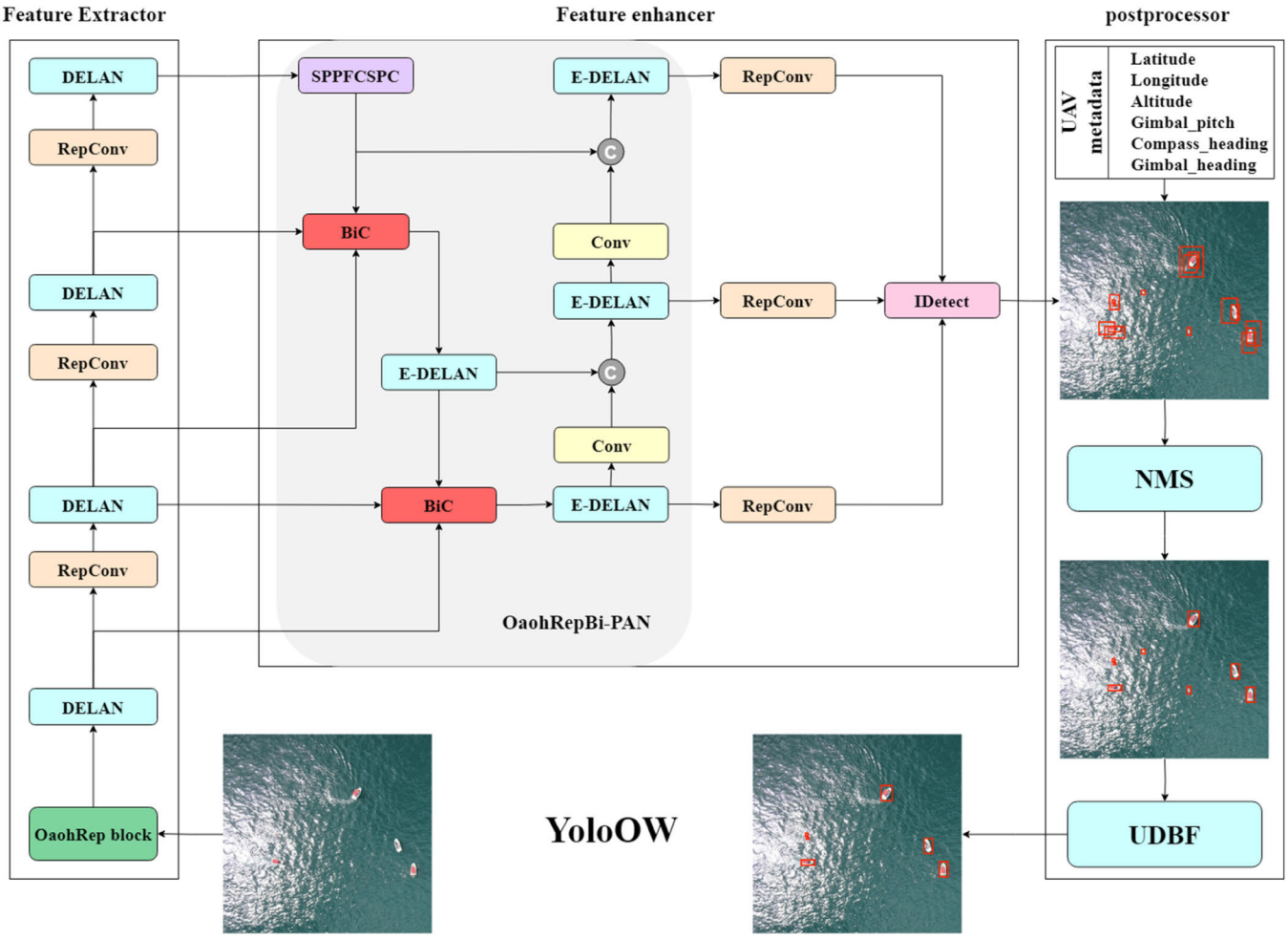


Fig. 3. Overall structure of YoloOW model. Given an original input image, the OaohRep block and DELAN generate feature maps at different and adaptive spatial scales and input them into the feature enhancer. Cross-layer and cross-scale feature fusion is achieved by using the OaohRepBi-PAN structure in the feature enhancer. These fused features are converted into bounding boxes by the IDetect module and input into the postprocessor. The final detection results are obtained by eliminating the redundant bounding boxes and false detection boxes caused by water surface flares using the NMS and UDBF module.

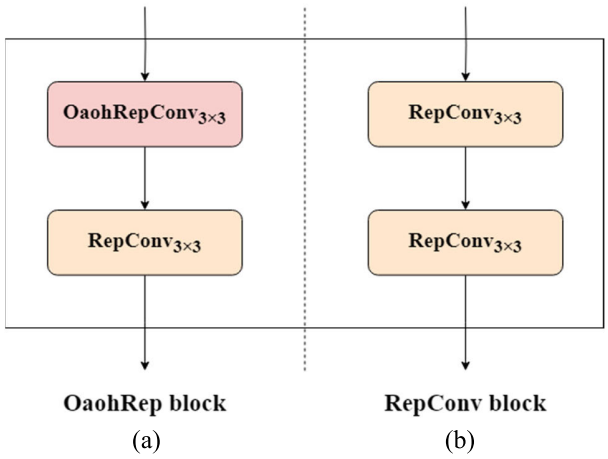


Fig. 4. (a) OaohRep block consists of a 3×3 size OaohRepConv and a 3×3 size RepConv. (b) RepConv block consists of a 3×3 size RepConv and a 3×3 size RepConv.

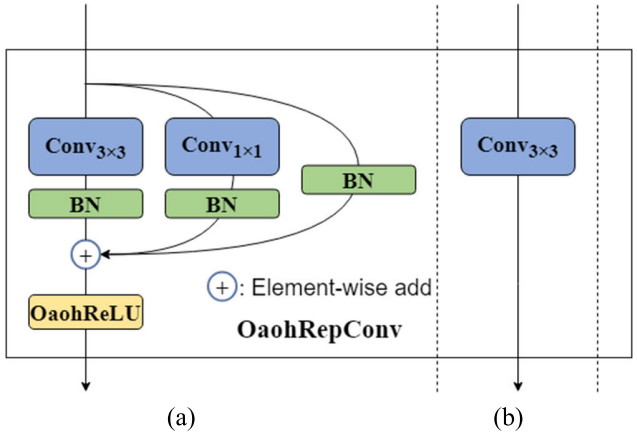


Fig. 5. (a) OaohRepConv consists of a 3×3 convolution kernel, a 1×1 convolution kernel, and an identity transformation matrix, with OaohReLU activation at training. (b) During detection, two convolution kernels and an identity transformation matrix are converted to a 3×3 convolution kernel and without activation function.

convolutional blocks, such as Visual Geometry Group (VGG) [47], leverage the advantages of high parallelism and a smaller memory footprint, resulting in higher inference speed.

Recently, some image classification models [48], [49] have attempted to decouple complex architectures during training from simplified architectures during inference, thus achieving

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an improved tradeoff between speed and accuracy, and RepConv block was proposed by Ding et al. [49], whose structure is illustrated in Fig. 4(b).

Inspired by these works, the RepConv convolutional layer has been utilized as the foundational layer within the OaohRep block. Based on RepConv, a novel convolution layer, named OaohRepConv, is proposed. Based on Fig. 5, OaohRepConv is comprised of multiple branches during the training process. This architectural design enables OaohRepConv to possess a highly adaptable receptive field while extracting features from imagery. When it comes to detection, the incorporation of weights acquired from various branches into a singular branch does not increase computational overhead.

The nonlinearity of neural networks stems from the utilization of activation functions. It is customary to insert a nonlinear activation function between two consecutive convolutional layers. The main idea of OaohReLU activation is to gradually reduce to an identity mapping with the increasing number of training epochs. During the training process, the two convolutions are gradually merged into one convolution. For OaohReLU activation $\mathrm{OaohReLU}(x)$, combine it with an identity mapping, which can be formulated as

$$\mathrm{OaohReLU}(x) = (1-\gamma)\mathrm{ReLU}(x) + \gamma x \tag{1}$$

where γ is a hyperparameter to balance the nonlinearity of the modified activation function $\mathrm{OaohReLU}(x)$. Denote the current epoch and the number of deep training epochs as e and E, respectively. We set $\gamma = (e/E)$. Therefore, at the beginning of training $(e = 0), \mathrm{OaohReLU}(x) = \mathrm{ReLU}(x)$, which means the network has strong nonlinearity. When the training converges, we have $\mathrm{OaohReLU}(x) = x$, which means the two convolutional layers have no activation functions in the middle.

Fig. 6 demonstrates the evolution of the OaohReLU activation function across training epochs. As the training progresses, the nonlinearity of the OaohReLU activation function gradually diminishes until it reaches a state of $\mathrm{OaohReLU}(x) = x$. During detection, $\mathrm{OaohReLU}(x) = x$ is also satisfied, and the cancellation of the activation function can effectively improve the speed of OaohRepConv.

When there is no activation function between two convolutional layers, these layers can be viewed as a single convolutional layer with a larger convolution kernel. Further demonstration will illustrate how to merge these two convolutional layers. The weights and bias matrix values of the convolution kernel can be ignored during the proof process, which are learned by the network.

Denote $a \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ and $b \in \mathbb{R}^{C_{\text{out}} \times H' \times W'}$ as the input and output features, the single convolution can be formulated as

$$b = W * a = W \cdot \mathrm{im2col}_{5\times 5}(a) \tag{2}$$

where $*$ denotes the convolution operation, $\cdot$ denotes the matrix multiplication, and W denotes the weight matrix of the convolution kernel. The im2col operation transforms the input into a matrix corresponding to the kernel shape. The two convolutions without activation function are

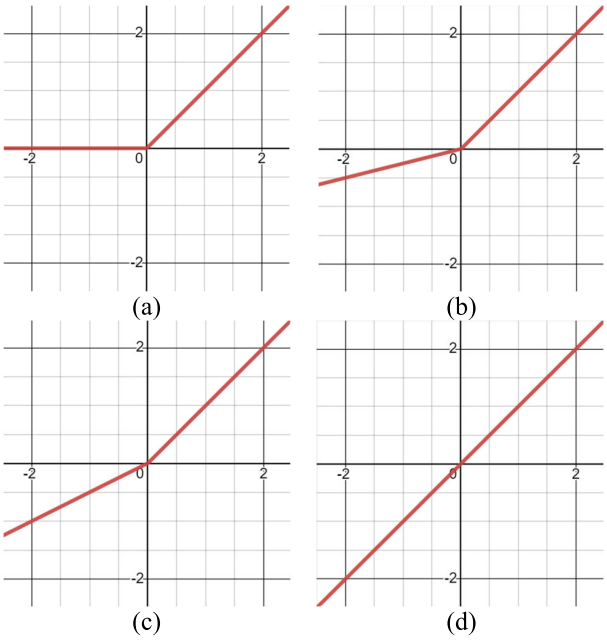


Fig. 6. (a)–(d) Variations of the OaohReLU activation function at the initiation of training, one-quarter progression, halfway through training, and upon the completion of the training process, respectively.

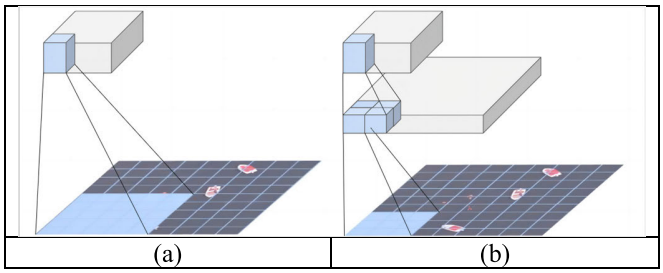


Fig. 7. Feature map representation of the OaohRep block. (a) End of the training. (b) Start of training.

formulated as

$$\begin{aligned} b &= W^1 * \left(W^2 * x\right) \\ &= W^1 \cdot \mathrm{im2col}_{3\times 3}\left(W^2 \cdot \mathrm{im2col}_{3\times 3}(a)\right) \\ &= W^1 \cdot W^2 \cdot \mathrm{im2col}_{3\times 3}(\mathrm{im2col}_{3\times 3}(a)). \end{aligned} \tag{3}$$

The im2col operation generates a matrix in which the number of columns is the size of the local receptive field of the convolution kernel, and the number of rows is the number of pixels in the output feature map. For two 3×3 convolution kernels, the receptive field is the same as a single 5×5 convolution kernel [47]. The number of pixels in the output feature map is also the same $(C_{\text{out}} \times H' \times W')$. Therefore, in the absence of activation functions and under the condition of neglecting differences in weight values, two 3×3 convolutional kernels can be merged as a single 5×5 convolutional kernel.

The OaohRep block begins training similarly to the RepConv block, consisting of two complete 3×3 RepConv layers, offering enhanced nonlinearity for robust learning capability within the model. However, the 3×3 RepConv exhibits limited receptive fields over the image, as depicted in Fig. 7(b). As the training progresses, in the later stages,

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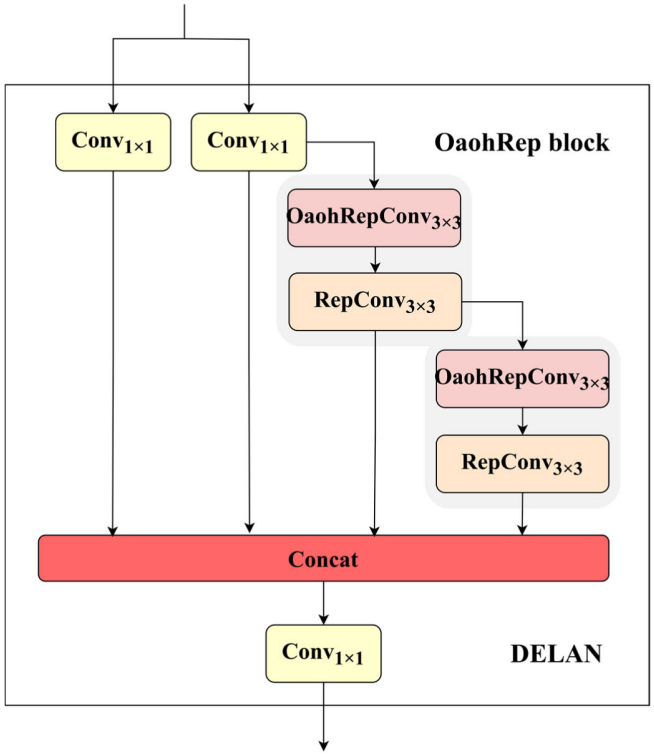


Fig. 8. Detailed structure of DELAN.

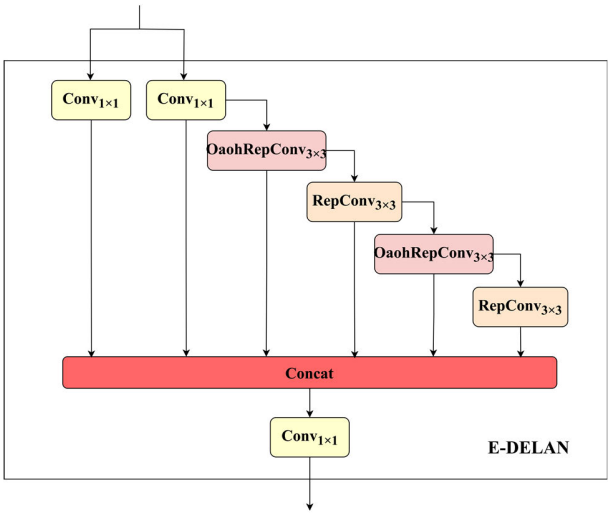


Fig. 9. Detailed structure of E-DELAN.

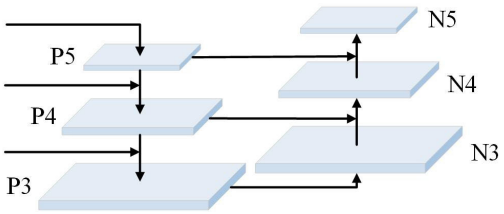


Fig. 10. Feature map structure of PANet.

the two convolutional layers, 3 × 3 OaohRepConv and 3 × 3 RepConv, within the OaohRep block are merged. As shown in Fig. 7(a), the OaohRep block achieves a broader receptive field over the image. Simultaneously, due to the reduction in model computations, the detection speed of the model is also enhanced.

B. Dynamically Adaptive Reparameterized Efficient Layer Aggregation Networks

For feature extractors, the accuracy is closely associated with the depth of the network [50]. However, as the network reaches a certain depth, the incremental gain in accuracy diminishes when additional calculation blocks are stacked [51], [52], [53]. Furthermore, at a critical depth, the convergence of the network starts to deteriorate, leading to overall accuracy that is inferior to shallower networks. As a state-of-the-art solution to address this issue, the efficient layer aggregation network (ELAN) [54] highlights that the diversity in network gradient aggregation might be the key concern.

The DELAN is an extension of the ELAN architecture, wherein the convolution block is substituted with the OaohRep block. By using OaohRep block, DELAN has two additional branch structures during training, which can increase the cardinality of the added features and combine the features of different groups in a shuffle and merge cardinality manner (Fig. 8). In the detection phase, the two convolutional layers in the OaohRep block structure will be merged into one convolutional layer (Fig. 7), which can effectively increase the speed while increasing the receptive field of the DELAN module.

Regardless of the gradient path length and the stacking number of computational blocks in DELAN, it has reached a stable state. If more computational blocks are stacked unlimitedly, this stable state may be destroyed, and the parameter utilization rate will decrease. Based on the gradient path analysis network design theory [54], E-DELAN is proposed to split the OaohRep block in DELAN to achieve more gradient path combinations. The proposed E-DELAN uses expand, shuffle, and merge cardinality to achieve the ability to continuously enhance the learning ability of the network without destroying the original gradient path (Fig. 9).

C. OaohRepBi-PAN

The PANet network [55], used in the feature enhancer of the general detector, achieves feature fusion at two scales in adjacent stages. As shown in Fig. 10, the P3 and P4 layer feature maps are obtained by fusing two adjacent features of this layer and the next layer.

However, it lacks fusions of low-level features. Numerous studies have demonstrated that low-level feature maps encompass many intricate features [56], [57], which is significant for detecting diminutive objects. The general solution is to add a P2 layer [37] based on PANet. Its limitation lies in increasing the complexity of the network and reducing the speed of object detection to a certain extent. We have observed a remarkable convergence between deep learning-based object detection models and traditional scale invariant feature transform (SIFT) algorithms in the design of feature extractors and feature enhancers. The SIFT algorithms possess scale-invariant

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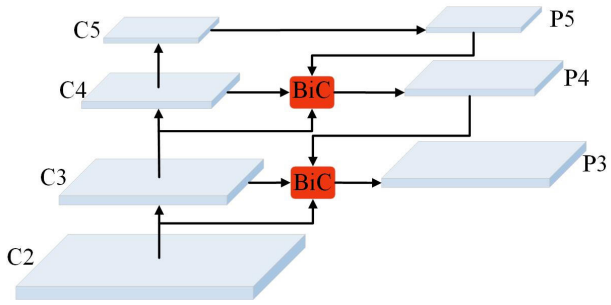


Fig. 11. Feature map structure of OaohRepBi-PAN.

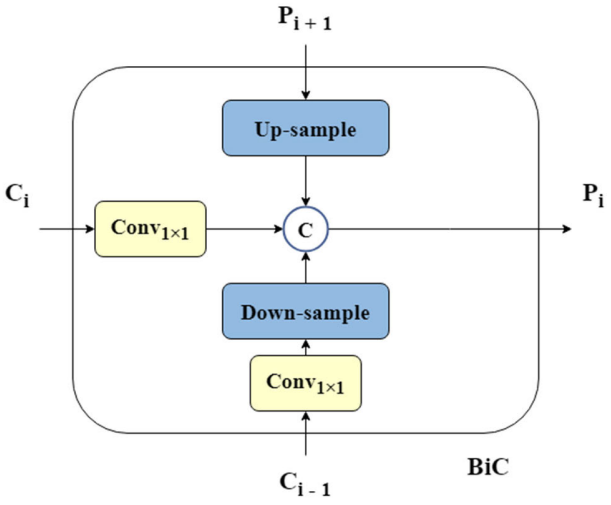


Fig. 12. Structure of a BiC module.

characteristics by constructing a Gaussian difference pyramid and computing the maximum values between adjacent upper, lower, and current feature layers to locate image features. Through observation, it has been noted that compared to PANet, the SIFT algorithm achieves scale invariance by additionally considering low-level features. Inspired by the SIFT operator, as illustrated in Fig. 11, the features extracted by the YoloOW feature extractor from the upper, lower, and current layers are fused using the bidirectional concatenation (BiC) module.

In OaohRepBi-PAN, the BiC module is proposed to integrate the feature maps of three adjacent layers, which fuses an extra low-level feature from the backbone C_{i-1} into P_i (Fig. 12).

D. UAV Detection Box Filter

The general detector predicts the detection results from the feature enhancer's output. However, solving the challenge of water surface flares in UAV aerial imagery of open water requires that the detector can postprocess [58], [59], [60] the feature enhancer's output. The difficulty with this postprocessing is that the existing general detectors do not have such a postprocessing module for water surface flares, and it is easy to produce false detection boxes for water surface flares [61]. Therefore, a detector with the ability to tackle water surface flares is urgently needed. After analyzing the output of the general detector, it was observed that the false detection boxes generated for most water surface flares exhibit a disparity in size compared to the true detection boxes of the actual categories. An immediate observation yields can filter out false detection boxes based on their sizes.

In essence, UDBF consists of three parts (Fig. 13).

The distance calculation module was proposed to calculate the relative distance $\boldsymbol{D}_i^{\mathbf{oc}}$ based on the UAV metadata and detection box coordinates.

The gradient boosting regression (GBR) fitting module is designed to learn the correlation between the relative distance $\boldsymbol{D}_i^{\mathbf{oc}}$ and the size of the detection box $\boldsymbol{S}_i^{\mathbf{oc}}$. It provides a prediction of the detection box size $\mathbf{Pre}\boldsymbol{S}_i^{\mathbf{oc}}$ based on the relative distance $\boldsymbol{D}_i^{\mathbf{oc}}$ during the validation and detection process.

The mean square-error regression (MSR) filtering module was proposed to compute the tolerance value range $\boldsymbol{T}^{\mathbf{oc}}$ of the GBR fitting module during validation. It is then utilized to filter out false detection boxes caused by water surface flares during detection.

1) Distance Calculation: The underlying 3-D geometric structure information of the scene is often ignored in object detection tasks. By leveraging the metadata provided by the UAV, it is feasible to reconstruct the mathematical perspective projection camera model, which resembles the typical use case for cameras on UAVs.

An estimation of the horizontal distance from the UAV to the object on the water is represented as y (refer to Fig. 14). In this estimation, assume a flat-water surface, disregarding any variations in elevation and the curvature of the Earth. This assumption is considered reasonable for UAV open water search and rescue missions that involve typical altitudes [62].

Using the focal length f and the y-axis pixel position offset v from the horizontal center pixel line of the image, compute y by computing the angle α via.

This allows us to compute y via

$$\alpha = \tan^{-1}\left(\frac{v}{f}\right) \tag{4}$$

$$y = \tan\left(90^\circ - (\beta - \alpha)\right)z. \tag{5}$$

Based on the provided image, determine the horizontal ground distance denoted as x by observing from a top–down perspective, as indicated in the lower left section of Fig. 14. To obtain this value, we require the variables l and w, which can be readily computed using Pythagoras' theorem. Subsequently, these variables enable us to calculate the value of x

$$l = \sqrt{z^2 + y^2}, \quad w = \sqrt{f^2 + v^2}, \quad x = \frac{u}{w}l. \tag{6}$$

Finally, calculate the relative distance d from the water surface object to the UAV through l and x

$$d = \sqrt{l^2 + x^2}. \tag{7}$$

2) GBR Fitting: GBR [63], a regression algorithm based on ensemble learning, is characterized by high predictive accuracy and robustness to outliers. The GBR fitting module is based on the GBR ensemble learning model, consisting of five key parameters: n_estimators, max_depth, and min_samples_split. The GBR model is composed of various base estimators

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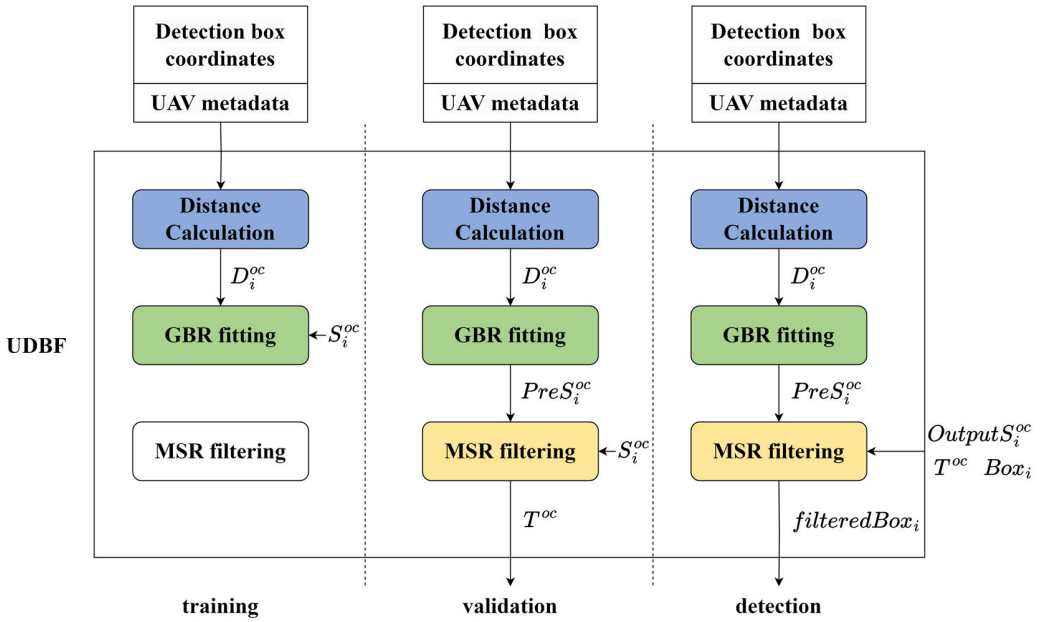


Fig. 13. Flowchart of UDBF. The UDBF and YoloOW models share a similar framework, which consists of three phases: training, validation, and detection. Each phase is associated with a different subset of data. During the training phase, the data are used to train the GBR fitting module within the UDBF. In the validation phase, the data are used to evaluate the effectiveness of the GBR fitting module. The MSE filtering module computes the tolerance value range $\boldsymbol{T}^{\mathbf{oc}}$ for the GBR fitting module during this phase. In the detection phase, the MSE filtering module uses $\boldsymbol{T}^{\mathbf{oc}}$ to filter out false detection boxes that fall outside the permissible error range. This is done by comparing the predicted detection box sizes $\mathbf{PreS}_{\boldsymbol{i}}^{\mathbf{oc}}$ generated by the GBR fitting module with the actual detection box sizes $\mathbf{Output}\boldsymbol{S}_{\boldsymbol{i}}^{\mathbf{oc}}$ outputted by the YoloOW model.

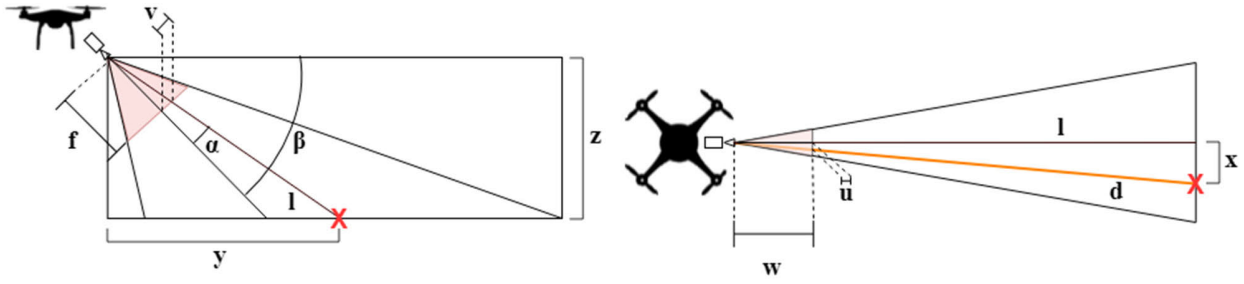


Fig. 14. Three-dimensional geometric structure of the mathematical perspective projection camera model. The provided figure illustrates the 3-D geometric correlation among various parameters, including the distance d (depicted by the orange line) from the drone to the object (represented by the red cross).

that form a random forest structure. The number of trees in the forest, or base estimators, is known as n_estimators. The maximum depth of a tree is called max_depth. The min_samples_split parameter represents the minimum number of samples required to split internal nodes. The accuracy of the random forest model is affected by n_estimators in a monotonic way. This means that the larger the value of n_estimators, the better the model performs. However, there is a limit beyond which increasing n_estimators do not improve the accuracy of the random forest. The max_depth and min_samples_split parameters help prevent overfitting issues that may arise from large amounts of sample data. The GBR fitting module is trained on the size $\boldsymbol{S}_{\boldsymbol{i}}^{\mathbf{oc}}$ and distance $\boldsymbol{D}_{\boldsymbol{i}}^{\mathbf{oc}}$ of each object's detection box in the training dataset. It provides a prediction of the detection box size $\mathbf{PreS}_{\boldsymbol{i}}^{\mathbf{oc}}$ based on the relative distance $\boldsymbol{D}_{\boldsymbol{i}}^{\mathbf{oc}}$ during the validation and detection process. The GBR fitting module is documented in Algorithm 1.

3) MSE Filtering: The mean square error (MSE) is used to verify the effectiveness of the GBR fitting module and calculates the tolerance value range $\boldsymbol{T}^{\mathbf{oc}}$ for the GBR fitting module in the validation phase. In the detection phase, the MSE filtering module uses $\boldsymbol{T}^{\mathbf{oc}}$ to filter out false detection boxes that fall outside the permissible error range. The MSE filtering module is documented in Algorithm 2.

IV. Experiments and Evaluations

A. Datasets

Robustness to scale variation and water surface flares are a requirement for an ideal detector in open water search and rescue scenarios, i.e., not only adapt to the scale variation of the UAV imagery but also reduce the interference of water surface flares. The article adopts a dataset: SeaDronesSee [22] for the complete evaluation of open water search and rescue scenes. Table I lists the relevant parameters of the dataset in the experiments and other widely used UAV object detection datasets. The comparison shows that the SeaDronesSee dataset, as the latest open water search and rescue dataset, not only has a wide altitude and angle range but also provides metadata of UAVs.

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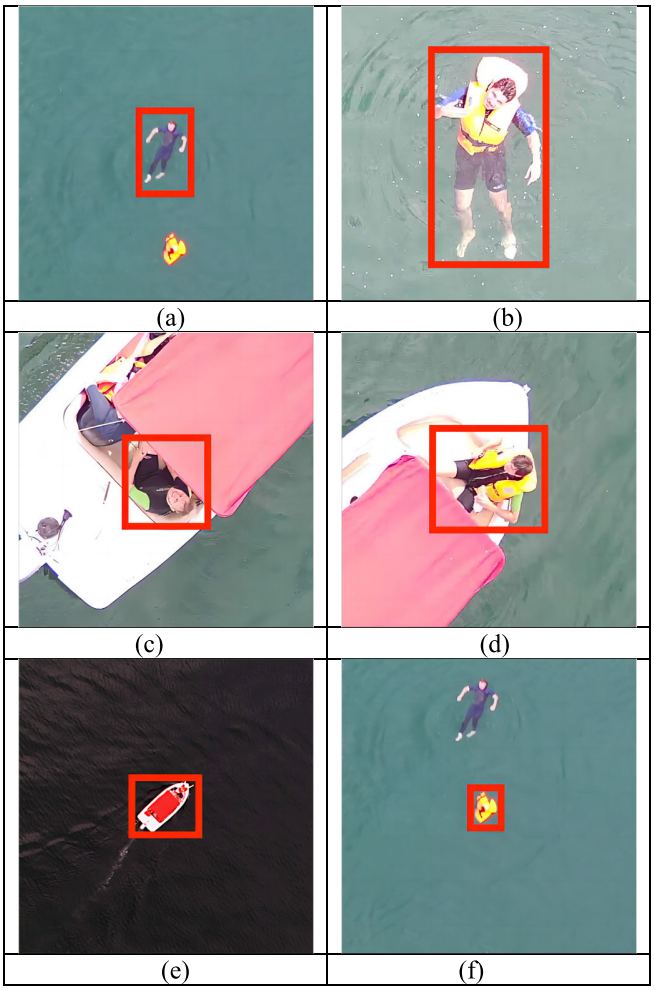


Fig. 15. Image of six predefined fine-grained categories. (a) Swimmer. (b) Floater. (c) Swimmer on a boat. (d) Floater on boat. (e) Boat. (f) Life jacket.

Algorithm 1 Gradient Boosting Regression Fitting

Input: $\boldsymbol{D_i^{oc}}$: the set of relative distances, where $\boldsymbol{D_i^{oc}} = \{\boldsymbol{d_i^{oc}} \in \mathbb{R}^+ | \boldsymbol{i = 1, 2, \ldots, N}\}$, $\mathbf{oc}$ = swimmer, floater, …, life jacket. $\boldsymbol{S_i^{oc}}$: the set comprising the size of detection boxes.

Initialization:

GBR {
"n_estimators": 500,
"max_depth": 4,
"min_samples_split": 5,
"learning_rate": 0.01,
"loss": "squared_error",
}

Fitting:

GBR.fit($\boldsymbol{D_i^{oc}}$, $\boldsymbol{S_i^{oc}}$)

Prediction:

$\boldsymbol{PreS_i^{oc}}$ = **GBR.predict**($\boldsymbol{D_i^{oc}}$)

The SeaDronesSee dataset contains images acquired by the UAV at different angles and altitudes, consisting of 2975 training images, 859 validation images, and 1796 testing images. The resolution of the imagery is distributed between 5456 × 3632 and 1280 × 960. The dataset provides bounding boxes

Algorithm 2 Mean Square Error Filtering

Validation:

Input: $\mathbf{PreS_i^{oc}}, \boldsymbol{S_i^{oc}}$

Output:

$\boldsymbol{T^{oc}}$ = **MSR**($\boldsymbol{S_i^{oc}}, \boldsymbol{PreS_i^{oc}}$)

Detection:

Input: $\boldsymbol{PreS_i^{oc}}, \boldsymbol{OutputS_i^{oc}}, \boldsymbol{T^{oc}}$ *and* $\boldsymbol{Box_i}$, where $\boldsymbol{Box_i}$ denotes the set of detection boxes.

Filtering:

IF MSR($\boldsymbol{OutputS_i^{oc}}, \boldsymbol{PreS_i^{oc}}$) > $\boldsymbol{T^{oc}}$ do

Delete $\boldsymbol{Box_i}$

Output: $\boldsymbol{filteredBox_i}$

TABLE I
COMPARISON WITH THE MOST PROMINENT ANNOTATED AERIAL DATASETS

Dataset	Env.	Altitude	Angle	Meta
DOTA	cities	Unknow	90°	×
UAVDT	traffic	5-200m	$0-90^\circ$	×
VisDrone	traffic	5-200m	$0-90^\circ$	×
Airbus Ship	maritime	Unknow	90°	×
SeadroneSee	maritime	5-260m	$0-90^\circ$	√

of instances and six predefined fine-grained category labels (boat, swimmer, floater, and so on) for training images and validation images manually annotated (Fig. 15). The SeaDronesSee dataset provides a test set of images without manual annotations. The accuracy of all models in experiments on the test set is reported on the official website to ensure fairness and impartiality. Furthermore, the dataset encompasses additional metadata for UAV imagery, such as GPS and inertial measurement unit (IMU) information, and the UAV metadata is the basis for the UDBF of our proposed YoloOW. All approach is trained on the training set and evaluated on the testing set, and ranking can be viewed on the website list.

B. Evaluation Metrics

This article mainly takes three criteria for different purposes.

Similar to the most relevant object detection works [64], [65], [66], AP(mean averaged precision), AP_{75}(Aps at the inertial measurement unit (IMU) threshold of 0.75), and AP_{50}(APs at the IoU threshold of 0.5) are used to evaluate our approach on the overall datasets, which are the most popular evaluation metrics for object detection. Since AP is a comprehensive indicator of precision and recall, the AP value calculated for the whole data reflects the detection performance for the overall dataset. They are the most used metrics in other UAV image object detection works. Larger AP and AP_{50} and AP_{75} indicate better performance.

AR_1 (the number of detection boxes is 1) and AR_{10} (the number of detection boxes is 10) are also commonly utilized metrics [5], which are the recall ratio corresponding to a fixed number of boxes in the image. AR_1 and AR_{10} reflect whether the detector can recall more instances. Larger AR_1 and AR_{10} indicate better performance.

In addition to the detection precision, running speed is another important performance metric for target detection

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algorithms. A common metric for evaluating speed is FPS, which represents the number of images that can be processed per second. Larger FPS indicates better performance.

C. Implement Details

1) *Baseline Model:* Considering the real-time constraints imposed by the algorithm, the baseline model is constrained to a one-stage target detection model. Yolov7 serves as the baseline model, which is presently regarded as a prominent one-stage target detection model in the research community. Yolov7 has achieved a balance between model accuracy and speed, demonstrating excellent real-time performance while ensuring model accuracy. In contrast to the currently popular decoupled head designs, both Yolov7 and YoloOW adopt a coupled head structure. In this article, Yolov7 was chosen as the baseline model for conducting ablation experiments.
2) *Train Phase:* In the training of SeaDronesSee dataset, 2975 images from the training set were used, while 869 images from the validation set and 1796 images from the test set were excluded from the training process. Consistent data augmentation techniques were applied to all models during training, including hue, saturation, value (HSV) color space adjustments, image geometric transformations, and random image augmentation. During training, the input image size was uniformly set to 1280 × 1280, which was influenced by the resolution sizes of images within the dataset and was consistent across all models. To ensure a fair comparison, all models underwent 300 epochs of training using the stochastic gradient descent (SGD) optimizer without utilizing pretrained weights, ensuring that all models could achieve optimal accuracy. The training weights that performed best on the validation set were used during the testing phase to prevent overfitting.
3) *Test Phase:* In the test on the SeaDronesSee dataset, the input image size was uniformly set to 1280 × 1280, which was the same as the training phase. The NMS threshold and maximum detection per image for all models were preset to 0.3 and 500, respectively, while the IoU threshold was set to 0.3. These parameters are influenced by the distribution and similarity of objects in the dataset and are consistent across all models.

D. Comparative Evaluation With State of the Arts

In Table II, a quantitative comparison between other state-of-the-art one-stage models and the YoloOW model is conducted on the SeaDronesSee dataset test set. Consistent experimental conditions were applied to all models. Each model underwent five repeated experiments, and the error margins are indicated in Table II. For the Yolov5 and Yolov8 models, although the number of parameters and FLOPS of the YoloOW model is lower than theirs, our detection performance is better than theirs. Compared with the Yolov7 [67] model and the Yolov7-sea [21] model improved for maritime search and rescue, YoloOW has achieved improvements in accuracy and speed by adding a small number of parameters.

TABLE II
ACCURACY COMPARISON WITH THE STATE-OF-THE-ART ONE-STAGE MODELS ON THE TEST SET OF THE SEADRONESSEE DATASET. CONSISTENT EXPERIMENTAL CONDITIONS WERE APPLIED TO ALL MODELS. "CN" STANDS FOR CENTERNET

Model	AP	AP_{50}	AP_{75}	AR_1	AR_{10}
Yolov5	28.16 ±0.07	49.48 ±0.13	27.59 ±0.18	15.77 ±0.08	34.79 ±0.47
Yolov7	28.75 ±0.09	49.23 ±0.18	28.75 ±0.24	16.19 ±0.08	36.01 ±0.38
Yolov8	33.47 ±0.13	56.01 ±0.21	34.85 ±0.23	20.09 ±0.09	41.11 ±0.41
Yolov7-sea	29.21 ±0.06	50.64 ±0.13	29.22 ±0.18	17.47 ±0.07	37.85 ±0.58
CN-Hourglass104	25.60 ±0.14	50.32 ±0.23	22.23 ±0.27	17.73 ±0.11	40.10 ±0.46
CN-ResNet101	15.12 ±0.07	36.42 ±0.11	10.82 ±0.14	9.62 ±0.06	21.44 ±0.43
CN-ResNet18	9.92 ±0.05	21.84 ±0.13	9.00 ±0.18	7.23 ±0.07	19.70 ±0.48
EffjcientDet-D0	20.80 ±0.07	37.11 ±0.12	20.62 ±0.19	11.51 ±0.09	29.10 ±0.50
YoloOW	**37.18** ±0.08	**65.39** ±0.17	**36.75** ±0.21	**23.17** ±0.08	**45.35** ±0.28

TABLE III
ACCURACY COMPARISON WITH THE STATE-OF-THE-ART TWO-STAGE MODELS ON THE TEST SET OF THE SEADRONESSEE DATASET. CONSISTENT EXPERIMENTAL CONDITIONS WERE APPLIED TO ALL MODELS. "F" STANDS FOR FASTER R-CNN. "C" STANDS FOR CASCADE R-CNN. "S" STANDS FOR SPARSE R-CNN

Model	AP	AP_{50}	AP_{75}	AR_1	AR_{10}
F.ResNeXt101FPN	30.40 ±0.12	54.66 ±0.16	29.39 ±0.21	18.58 ±0.07	42.55 ±0.57
F.ResNet50 FPN	14.20 ±0.07	30.09 ±0.12	12.45 ±0.16	9.43 ±0.05	25.64 ±0.51
C.ResNet101 FPN	24.14 ±0.06	37.77 ±0.09	26.10 ±0.17	12.95 ±0.07	28.92 ±0.39
C.ResNet50 FPN	24.10 ±0.10	37.96 ±0.14	25.95 ±0.19	13.01 ±0.08	29.11 ±0.48
S.ResNet101 FPN	28.34 ±0.07	51.62 ±0.13	27.20 ±0.21	18.71 ±0.06	45.65 ±0.28
S.ResNet50 FPN	26.30 ±0.09	48.05 ±0.16	24.78 ±0.19	19.59 ±0.08	45.37 ±0.47
YoloOW	**37.18** ±0.08	**65.39** ±0.17	**36.75** ±0.21	**23.17** ±0.08	**45.35** ±0.28

Compared with the baseline model Yolov7, YoloOW improved the AP index by 8.43%. Compared to other one-stage networks, such as CenterNet [23] and EfficientDet [30], YoloOW has exhibited an advantage in accuracy. For conducting sufficient comparative experiments, CenterNet used three different backbone networks with increasing complexity.

In Table III, a quantitative comparison between other state-of-the-art two-stage models and the YoloOW model is conducted on the SeaDronesSee dataset test set. Consistent experimental conditions were applied to all models. Each model underwent five repeated experiments, and the error margins are indicated in Table III. Compared to two-stage detectors, such as Faster R-CNN [68], Cascade R-CNN [7], and Sparse R-CNN [69], our proposed YoloOW performs

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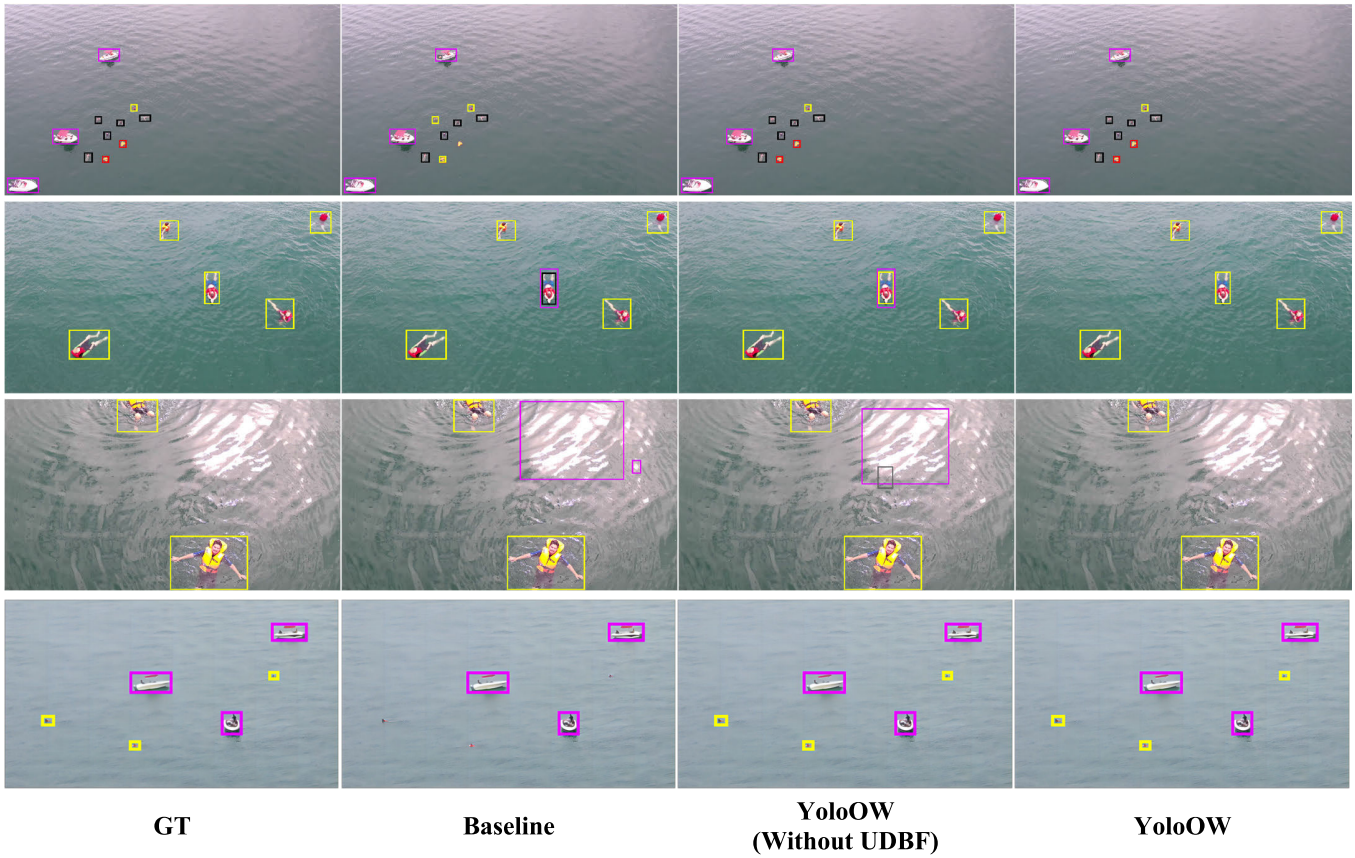


Fig. 16. Visualization of detection results among ground truth, the baseline of Yolov7, YoloOW (without UDBF), and YoloOW on the SeaDronesSee dataset test set.

better than these two-stage methods. The accuracy of YoloOW implementation of AP 37.18% is higher than the best accuracy of these three methods: 30.40%, 24.14%, and 28.34%. Moreover, the two-stage detectors exhibit significantly slower detection speeds than their one-stage counterparts.

E. Visualization Analysis

The qualitative comparison of detection visualization results among the baseline model (Yolov7), YoloOW (without UDBF), and our proposed YoloOW is shown in Fig. 14. In the first image, the baseline model not only confuses the "lifejacket" and "floater" categories, falsely detecting "lifejacket" as "floater," but also confuses the "floater" and "swimmer" categories, falsely detecting "swimmer" as "floater." In the first image, the YoloOW detection results are consistent with ground truth, showing excellent classification performance. It is observed that the YoloOW model outperforms other detection models in all fine-grained categories, especially in confusing categories, such as "swimmer," "floater," and "life jacket." In the second and third images, the baseline model and YoloOW (without UDBF) produce false detection boxes for the "boat" category, which are purple detection boxes on the image. The visualization results show the efficacy of UDBF in eliminating false detection boxes attributed to water surface flares, such as the filtered purple detection boxes in the second and third rows in Fig. 16. In the fourth image, the baseline model misses the detection of the tiny "floater" category in the image. The YoloOW model demonstrates commendable recall performance and effectively alleviates the issue of missed detection in small targets present in the baseline model.

TABLE IV

ACCURACY COMPARISON AMONG THE BASELINE MODELS OF YOLOV7 AND MODELS WITH DIFFERENT COMPONENTS IN OUR APPROACH ON THE TEST SET OF SEADRONESSEE DATASET. "FEX" STANDS FOR FEATURE EXTRACTOR. "FEN" STANDS FOR FEATURE ENHANCER. "PP" STANDS FOR POSTPROCESSOR

FEX	FEN	PP	AP	AP_{50}	AP_{75}	AR_1	AR_{10}
			28.75	49.23	28.75	16.19	36.01
√			33.98	57.02	34.87	21.34	42.12
√	√		37.16	65.32	36.75	23.17	45.35
√	√	√	37.18	65.39	36.75	23.17	45.35

F. Ablation Studies

To verify the contribution of each component in our proposed approach, this section conducts thorough ablation experiments on the test set of the SeaDronesSee dataset, and the baseline is fixed at Yolov7. The accuracy results of the ablation experiments in Table IV are for the whole instances in the dataset, while the visualization results of ablation experiments are shown in Fig. 16. The accuracy results of the ablation experiments for the fine-grained categories are shown in Fig. 17.

Fig. 16 shows the visualization detection result of the ablation experiments by adding UDBF based on the YoloOW.

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TABLE V
COMPARISON OF MODEL SPACE COMPLEXITY AND TIME COMPLEXITY WITH OTHER STATE-OF-THE-ART MODELS. CONSISTENT EXPERIMENTAL CONDITIONS WERE APPLIED TO ALL MODELS

Model	AP	Params	FLOPs	FPS
Yolov5	28.16	46.1	107.7	37.8
Yolov7	28.75	**36.5**	103.3	58.8
Yolov8	33.47	43.6	165.4	30.2
Yolov7-sea	29.21	36.9	116.5	49.0
YoloOW	**37.18**	39.3	**88.0**	**65.3**

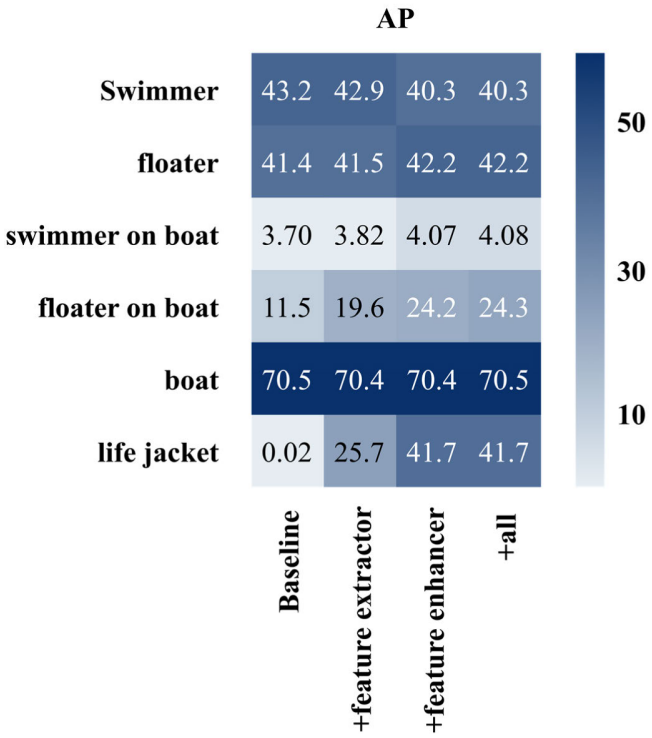


Fig. 17. Ablation studied six fine-grained categories with AP on the SeaDronesSee test set. Gradually replace the feature extractor, feature enhancer, and postprocessor based on the Yolov7 model. "All" indicates that all three components are added. The darker the color, the higher the accuracy.

Although after quantitative analysis, the UDBF has not significantly improved the detection accuracy, the visual results show that it effectively removes false detection boxes caused by water surface flares.

According to Table IV, replacing the feature extractor in the baseline model with the feature extractor of the YoloOW model has significantly improved the accuracy and recall rates. The AP has increased from 28.75% to 33.98%, and the AR_1 has increased from 16.19% to 21.34%. Furthermore, by replacing the feature enhancer of the baseline model with the feature enhancer of YoloOW, the model's accuracy and recall rates continued to improve. The AP has increased from 33.98% to 37.16%, and the AR_1 has increased from 21.34% to 23.17%. However, the increase in the UDBF did not result in a substantial enhancement in detection accuracy. Nevertheless, based on the visualization results, it can be observed that the false detection boxes attributed to water surface flares were effectively eliminated.

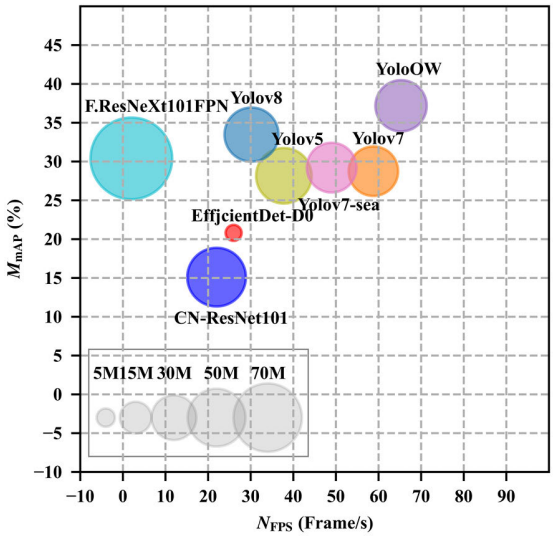


Fig. 18. Scatter plot about AP-FPS-Params of YoloOW and other state-of-the-art models in V100 device.

Fig. 17 shows that our feature extractor and feature enhancer can improve AP in five categories, especially the two categories of "life jacket" and "floater on the boat," which increased by 41.5% and 12.8%, respectively. This is a remarkable breakthrough as accurately detecting "life jacket" in this dataset is challenging, and it holds significant importance in open water rescue operations. Due to the scale variation of UAV images in the SeaDronesSee dataset, many detectors incorrectly detect "life jacket" as "floater."

Based on the observations from Fig. 17, it can be noted that the incorporation of the UDBF has led to a marginal enhancement in the accuracy of the three distinct categories, which are "boat," "floater on the boat," and "swimmer on the boat." This finding indicates that the false detection categories resulting from water surface flare predominantly pertain to the aforementioned categories. This result is consistent with the visualization results, proving the effectiveness of the UDBF.

G. Space Complexity and Time Complexity Analysis

A quantitative comparison was performed between other state-of-the-art models and YoloOW in terms of space complexity and time complexity. Params (M) represents the aggregate count of learnable parameters within the model, signifying the model's size. The smaller Params (M) means the model is lighter. FLOPs (B) are the total number of floating-point operations during the model's detection process. It is a metric of the computational complexity of the model and can be used to evaluate the model's time complexity. The smaller FLOPs (B) mean the model is faster.

The Params and FLOPs of YoloOW are 39.3 M and 88 B, which is even reduced by 4.3 M and 77.4 B compared to the model Yolov8 with the next best detection accuracy of 43.6 M and 165.4 B. In comparison to the most lightweight model, the Yolov7 baseline model, the YoloOW model achieved an improvement of 8.43% and 15.3 B in AP and FLOPs, while only introducing an additional 2.8 M Params (Table V).

In Fig. 18, the size of each circle corresponds to the magnitude of the model parameters. A larger circle indicates

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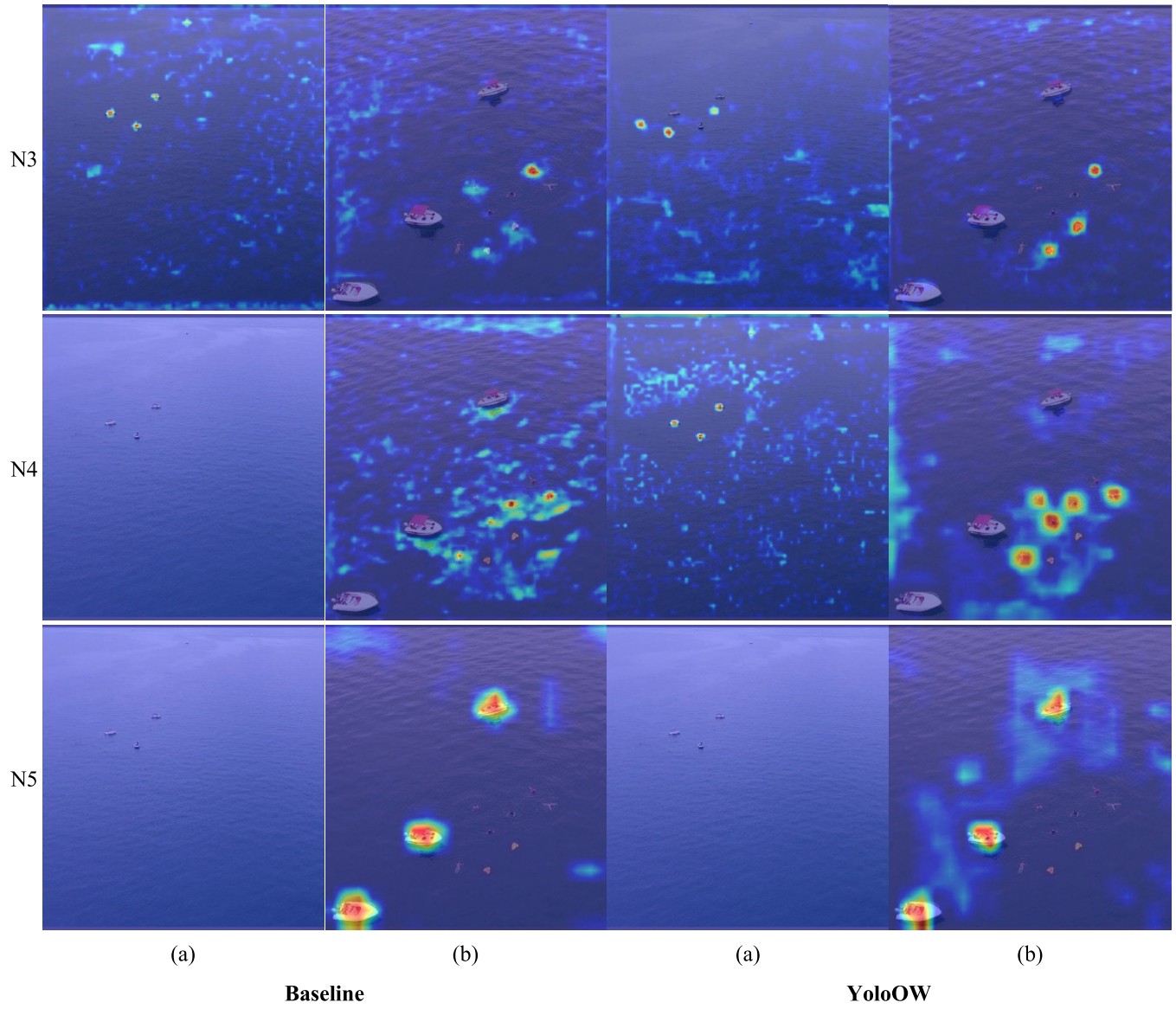


Fig. 19. Grad-CAM visualization results of baseline and YoloOW model on N3, N4, and N5. (a) Image with tiny objects. (b) Image with easily misidentified objects.

a higher number of model parameters. FPS is a metric that provides a direct measurement of the model's speed. A higher FPS implies a faster speed of the model. The YoloOW model achieves an impressive frame rate of 65.3 FPS, surpassing the real-time requirement of 30 FPS. The scatter plot visually demonstrates the advantage of the YoloOW model in terms of speed, accuracy, and model parameters.

H. Grad-CAM Map Analysis

Gradient-weighted class activation mapping (Grad-CAM) [70] is a visualization method that uses gradients to calculate the importance of spatial positions in the image. The Grad-CAM maps were generated from the N3, N4, and N5 of the baseline model and YoloOW. As shown in Fig. 10, it can be observed that N3, N4, and N5 correspond to the features extracted at various spatial scales from the image. The Grad-CAM graph generated by N3, N4, and N5 can qualitatively analyze the spatial scale adaptation performance of the YoloOW model.

Two classic images (a) and (b) in the SeaDronesSee test set were used to create Grad-CAM visualization maps. Image (a) has two categories "boat" and "floater." In this image, the UAV is positioned at a considerable distance from the scene, resulting in a reduced scale of the imaged object. The "floater" category has only 15–37 pixels in the image. The detection performance of the model can be qualitatively analyzed based on the image (a)'s Grad-CAM visualization maps. In image (b), three categories can be easily confused: "swimmer," "floater," and "life jacket." The classification performance of the model can be qualitatively analyzed based on the image (b)'s Grad-CAM visualization maps.

In Fig. 19, image (a) demonstrates that the baseline model cannot pay attention to the tiny "floater" category, attention is solely on the "boat" category, while the YoloOW model shows attention to the "floater" category in the N3 Grad-CAM visualization map of image (a). In the N4 Grad-CAM visualization map, image (b) shows that the baseline model's attention is focused on the local location of the objects and

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the YoloOW model forms powerful and global attention to the objects. Compared to the baseline model, the YoloOW model demonstrates competitiveness in scale adaptability, classification performance, and localization accuracy.

V. Conclusion

In this article, we proposed a real-time object detection neural network named YoloOW that adapts to different spatial scales for improved object detection performance in open water search and rescue from UAV aerial imagery. Three new model architectures were proposed in the YoloOW model to improve scale adaptability: OaohRep block, DELAN and its enhanced variant E-DELAN, and OaohRepBi-PAN. Compared with general convolution blocks, the OaohRep convolution block can extract image features across a wide range of spatial scales, show better scale adaptability, and achieve faster detection speed due to its unique merged convolution layer design. A novel architecture termed DELAN and its enhanced variant E-DELAN were proposed to augment the learning capability during model training and diminish floating point operations in the model's inference phase. OaohRepBi-PAN imitated SIFT's architecture and was applied to deep learning models, displaying better scale adaptability than PANet. Furthermore, a novel UAV detection frame filtering module was proposed to alleviate false detections caused by water surface flares. The evaluations on the SeaDronesSee dataset demonstrate the effectiveness, robustness, and characteristics of our proposed approach. The YoloOW model underwent quantitative evaluation in accuracy, speed, and model size. It has outperformed all other models with 37.18 AP, securing the top position on the leaderboard. Additionally, it achieves a speed of 65.3 FPS, which is faster than any other model with similar accuracy. Moreover, the YoloOW model has only 2.8 M more parameters than the lightest model, Yolov7, in model size.

However, the YoloOW model exhibited suboptimal detection performance in scenarios characterized by dense occlusion. In the future, we aim to address the limitations of the YoloOW model and strive for improved detection performance in densely occluded scenes.

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